

Milestone 2 Report

Jan 2026 DLGenAI Project – Messy Mashup

Music Genre Classification under Noisy Conditions

1. Introduction

This milestone focuses on exploratory data analysis and baseline model development for the “Messy Mashup” music genre classification competition. The objective is to analyze the provided instrument-separated training dataset and build a simple machine learning model to classify music genres.

The dataset contains 10 different genres, with each song split into four stems (drums, vocals, bass, and other). The goal is to understand audio characteristics and evaluate a baseline Decision Tree classifier.

2. Dataset Overview

The training dataset consists of:

- 10 music genres:
 - Blues
 - Classical
 - Country
 - Disco
 - HipHop
 - Jazz
 - Metal
 - Pop

- Reggae
 - Rock
- Each genre contains 100 songs.
- Each song is split into 4 stems:
 - drums.wav
 - vocals.wav
 - bass.wav
 - other.wav / others.wav

Total songs in training set: 1000
Total stems: 4000

3. Exploratory Data Analysis (EDA)

3.1 Mean Duration of Jazz Stems

The average duration of Jazz genre stems was computed.

Result:

Mean Duration $\approx$ **30.03 seconds**

This confirms consistency in track length across the dataset.

3.2 Unique Sample Rates

The dataset was analyzed to identify unique sampling rates.

Result:

Unique Sample Rate: **[44100]**

This indicates all audio files use a consistent sampling rate.

3.3 Corrupted / Zero-Byte Files

The dataset was checked for corrupted or zero-size files.

Result:

Number of corrupted files = **0**

This confirms dataset integrity.

3.4 Average Peak Amplitude (Vocal Stems)

Peak amplitude (in dB) was calculated for vocal stems.

Formula used:

Peak dB = $20 \times \log_{10}(\max(|\text{signal}|))$

Result:

Average Peak Amplitude $\approx$ **-12.49 dB**

This indicates normalized and well-scaled audio signals.

3.5 Mean Spectral Centroid (Blues Genre)

Spectral centroid represents the brightness of audio.

Result:

Mean Spectral Centroid (Blues) $\approx$ **1597.84 Hz**

This shows moderate frequency concentration typical for blues music.

3.6 Genre with Highest Mean Spectral Centroid

All genres were analyzed for mean spectral centroid.

Result:

Highest Spectral Centroid Genre = **HipHop**

This suggests HipHop tracks contain stronger high-frequency components.

3.7 Silence in First 0.5 Seconds

Stem files were checked for silence in the first 0.5 seconds.

Result:

Number of stem files containing silence = **2**

This indicates minimal silent padding in the dataset.

4. Feature Extraction

For model training, the following features were extracted from the “other” stem:

- Tempo (BPM)
- Spectral Centroid
- Zero Crossing Rate (ZCR)
- Spectral Rolloff

Audio was loaded at:

- Sampling rate: 22050 Hz
 - Duration: First 10 seconds
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5. Model Development

A Decision Tree classifier was implemented with:

- max_depth = 5
- random_state = 42

Stratified train-validation split:

- 80% Training
 - 20% Validation
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6. Model Performance

6.1 Validation Macro F1 Score

Macro F1 Score $\approx$ **0.1523**

This indicates baseline-level performance due to limited features and noisy dataset conditions.

6.2 Precision of HipHop

Precision (HipHop) = **0.25**

6.3 Recall of Pop

Recall (Pop) = **0.20**

6.4 Model Accuracy

Accuracy $\approx$ **0.19**

This is slightly above random baseline (10 classes $\rightarrow$ 10% random accuracy).

6.5 Confusion Matrix Analysis

- Genre with Highest True Positives: **Metal**

- Genre with Lowest False Negatives: **Metal**

This indicates the model better distinguishes metal compared to other genres.

7. Observations

- Decision Tree performance is relatively low due to:
 - Limited feature set
 - Noisy mashup conditions
 - Distribution shift between training and test data
 - Metal and HipHop appear to have more distinct frequency characteristics.
 - Classical and Country appear more confused in classification.
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8. Conclusion

This milestone successfully:

- Performed detailed dataset analysis
- Verified dataset consistency
- Extracted relevant audio features
- Built a baseline Decision Tree classifier
- Evaluated performance using macro F1 score and confusion matrix

The baseline results indicate the need for:

- Advanced feature extraction (MFCC, Mel Spectrograms)

- Deep learning models (CNNs)
- Data augmentation
- Regularization techniques

These improvements will be explored in future milestones.